

MIT 18.06 - Problem Set 6

* Section 4.3

$$4) \quad A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 8 \\ 8 \\ 20 \end{bmatrix}, \quad x = \begin{bmatrix} C \\ D \end{bmatrix}$$

$$Ax - b = \begin{bmatrix} C \\ C + D - 8 \\ C + 3D - 8 \\ C + 4D - 20 \end{bmatrix}, \quad \text{and we have} \quad E = C^2 + (C + D - 8)^2 + (C + 3D - 8)^2 + (C + 4D - 20)^2$$

$$E = C^2 + (C + D - 8)(C + D - 8) + (C + 3D - 8)(C + 3D - 8) + (C + 4D - 20)(C + 4D - 20)$$

$$= C^2 + (C^2 + 2CD - 16C - 16D + D^2 + 64) + (C^2 + 6CD + 9D^2 - 16C - 48D + 64) + (C^2 + 16D^2 + 8CD - 40C - 160D + 400)$$

$$= 4C^2 + 16CD - 72C - 224D + 26D^2 + 528$$

$$\frac{\partial E}{\partial C} = 8C + 16D - 72$$

$$\frac{\partial E}{\partial D} = 16C - 224 + 52D$$

$$A^T A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 8 \\ 8 & 26 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ 8 \\ 8 \\ 20 \end{bmatrix}$$

$$= \begin{bmatrix} 36 \\ 112 \end{bmatrix}$$

So $A^T A \hat{x} = A^T b :-$

$$\begin{bmatrix} 4 & 8 \\ 8 & 26 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 36 \\ 112 \end{bmatrix}$$

Which gives:- $4C + 8D = 36 \rightarrow 4C + 8D - 36 = 0$
 $8C + 26D = 112 \rightarrow 8C + 26D - 112 = 0$

Each equation multiplied by 2 gives the equations we get by setting the derivation from earlier equal to 0.

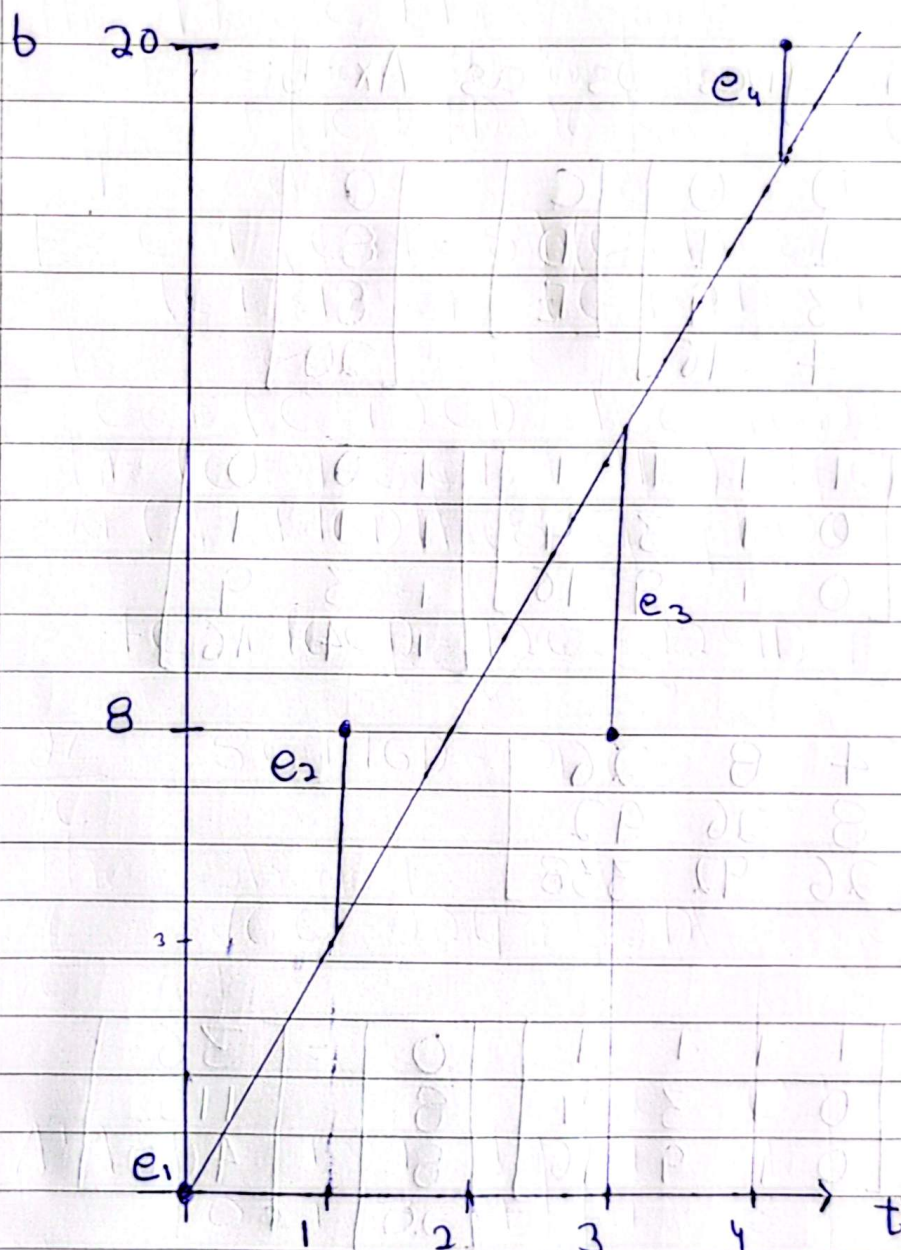
$$\neg) \begin{matrix} D(0) = 0 \\ D(1) = 8 \\ D(3) = 8 \\ D(4) = 20 \end{matrix} \Rightarrow A = \begin{bmatrix} 0 \\ 1 \\ 3 \\ 4 \end{bmatrix} \text{ and } b = \begin{bmatrix} 0 \\ 8 \\ 8 \\ 20 \end{bmatrix}$$

$$A^T A \hat{x} = A^T b$$

$$\begin{bmatrix} 0 & 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 3 \\ 4 \end{bmatrix} [\hat{D}] = \begin{bmatrix} 0 & 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ 8 \\ 8 \\ 20 \end{bmatrix}$$

$$26\hat{D} = 112$$

$$\hat{D} = 112/26 = 56/13$$



$$9) \quad 0 \neq C + D + E \neq 0$$

$$\frac{A \cdot C}{C + D + E} = 0$$

$$9) \quad \begin{aligned} C &= 0 \\ C + D + E &= 8 \\ C + 3D + 9E &= 8 \\ C + 4D + 16E &= 20 \end{aligned}$$

Which gives us as $Ax = b$:-

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{bmatrix} \begin{bmatrix} C \\ D \\ E \end{bmatrix} = \begin{bmatrix} 0 \\ 8 \\ 8 \\ 20 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 4 \\ 0 & 1 & 9 & 16 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 8 & 26 \\ 8 & 26 & 92 \\ 26 & 92 & 338 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 4 \\ 0 & 1 & 9 & 16 \end{bmatrix} \begin{bmatrix} 0 \\ 8 \\ 8 \\ 20 \end{bmatrix} = \begin{bmatrix} 36 \\ 112 \\ 400 \end{bmatrix}$$

So $A^T A x = A^T b$ is:-

$$\begin{bmatrix} 4 & 8 & 26 \\ 8 & 26 & 92 \\ 26 & 92 & 338 \end{bmatrix} \begin{bmatrix} \hat{C} \\ \hat{D} \\ \hat{E} \end{bmatrix} = \begin{bmatrix} 36 \\ 112 \\ 400 \end{bmatrix}$$

We have three vectors a_1, a_2 and a_3 (i.e. the first three columns of A), we have the vector b outside the span of the a 's, and we have $p = \hat{C}a_1 + \hat{D}a_2 + \hat{E}a_3$.

$$26) \quad A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \text{ and } x = \begin{bmatrix} C \\ D \\ E \end{bmatrix}$$

$$\text{Then } b - Ax = \begin{bmatrix} -C - D \\ 1 - C - E \\ 3 - C + D \\ 4 - C + E \end{bmatrix}$$

$$A^T(b - Ax) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} -C - D \\ 1 - C - E \\ 3 - C + D \\ 4 - C + E \end{bmatrix}$$

$$= \begin{bmatrix} -C - D \\ -C - D \\ 0 \end{bmatrix} + \begin{bmatrix} 1 - C - E \\ 0 \\ 1 - C - E \end{bmatrix} + \begin{bmatrix} 3 - C + D \\ -3 + C - D \\ 0 \end{bmatrix} + \begin{bmatrix} 4 - C + E \\ 0 \\ -4 + C - E \end{bmatrix}$$

Since $b - Ax$ is in $N(A^T)$, we equate the previous equation to 0 to get:

$$-C - D + 1 - C - E + 3 - C + D + 4 - C + E = 0$$

$$-C - D - 3 + C - D = 0$$

$$1 - C - E - 4 + C - E = 0$$

↓

$$-4C + 8 = 0$$

$$C = 2$$

$$-2D - 3 = 0$$

$$D = -3/2$$

$$-2E - 3 = 0$$

$$E = -3/2$$

The average of b 's is 8, and at $(0,0)$ we have:

$$C + D(0) + E(0)$$

$$= C = 2$$

29) First blank:- a_1 and a_2 are linearly dependent

$\emptyset$, $a_1, a_2, \dots, a_{n-1}, a_n$ will be contained in exactly one hyperplane if all the vectors $a_1, \dots, a_{n-1}$ are linearly independent.

Section 4.4

10) a) Suppose $c_1 q_1 + c_2 q_2 + c_3 q_3 = 0$.

Then dot producting both sides with q_1 gives $c_1 = 0$, both sides with q_2 gives $c_2 = 0$ and both sides with q_3 gives $c_3 = 0$.

b) Suppose $Qx = 0$ for some Q and x .

Then we have $Q^T Q x = 0$
 $I x = 0$
 $x = 0$

Thus only 0 is in $N(Q)$
therefore the columns of Q
are independent.

$$18) \quad A = a / \|a\|$$

$$= \frac{(1, -1, 0, 0)}{\sqrt{2}}$$

$$= (1/\sqrt{2}, -1/\sqrt{2}, 0, 0)$$

$$B = \frac{b - AA^T b}{\|b - AA^T b\|}$$

$$\neq b - AA^T b = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 \end{bmatrix}$$

$$\downarrow$$

$$\begin{bmatrix} 0 & 1 & -1 & 0 \end{bmatrix}^T$$

$$b - AA^T b = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1/2 & -1/2 & 0 & 0 \\ -1/2 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} - \begin{bmatrix} -1/2 \\ 1/2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/2 \\ -1 \\ 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 1/2 \\ 1/2 \\ -1 \\ 0 \end{bmatrix} \div \frac{\sqrt{6}}{2} = \begin{bmatrix} \sqrt{6}/6 \\ \sqrt{6}/6 \\ -\sqrt{6}/3 \\ 0 \end{bmatrix}$$

$$C = \frac{c - AA^T c - BB^T c}{\|c - AA^T c - BB^T c\|}$$

$$AA^T c = \begin{bmatrix} 1/2 & -1/2 & 0 & 0 \\ -1/2 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \text{ so } c - AA^T c = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

$$BB^T c = \begin{bmatrix} \sqrt{6}/6 \\ \sqrt{6}/6 \\ -\sqrt{6}/3 \\ 0 \end{bmatrix} \begin{bmatrix} \sqrt{6}/6 & \sqrt{6}/6 & -\sqrt{6}/3 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1/6 & 1/6 & -1/3 & 0 \\ 1/6 & 1/6 & -1/3 & 0 \\ -1/3 & -1/3 & 2/3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1/3 \\ -1/3 \\ 2/3 \\ 0 \end{bmatrix}$$

$$c - AA^T c - BB^T c = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix} - \begin{bmatrix} -1/3 \\ -1/3 \\ 2/3 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \\ -1 \end{bmatrix}$$

$$C = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \\ -1 \end{bmatrix} \div \frac{2\sqrt{3}}{3} = \begin{bmatrix} \sqrt{3}/6 \\ \sqrt{3}/6 \\ \sqrt{3}/6 \\ -\sqrt{3}/2 \end{bmatrix}$$

36
35) The n columns of Q_1 give us a basis for the column space of A . The $m-n$ columns of Q_2 give a basis for the left nullspace of A .

Section 5.1

10) $Ax=0$ is solved by $x = [1 \ 1 \ 1 \ \dots \ 1]^T$ and since $N(A)$ is non trivial its determinant is 0

$A-I$ subtracts 1 from each entry on the diagonal from each row of A , thus if the entries in every row previously added to 1, now they add to 0; therefore $\det(A-I)=0$ by the same logic as before.

It does not mean $\det(A) \neq 1$, e.g. let $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, it has $\det(A) = -1$ but $\det(A-I) = 0$

$$18) \begin{bmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & a & a^2 \\ 0 & b-a & b^2-a^2 \\ 0 & c-a & c^2-a^2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & a & a^2 \\ 0 & b-a & b^2-a^2 \\ 0 & 0 & (c^2-a^2)-(b+a)(c-a) \end{bmatrix}$$

$$\begin{aligned} \text{Determinant} &= b-a \left((c^2-a^2) - (b+a)(c-a) \right) \\ &= b-a \left((c+a)(c-a) - (b+a)(c-a) \right) \\ &= (b-a)(c-a) (c+a - b-a) \\ &= (b-a)(c-a)(c-b) \end{aligned}$$

~~31~~